

Operating Systems

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38. RAID

RAID (Redundant Array of Inexpensive Disks)

- ▣ **RAID** is to use multiple disks to build **faster, bigger, and more reliable** disk system.
- ▣ RAID is arranged into six different levels.
 - ◆ RAID Level 0: Striping multiple disks
 - ◆ RAID Level 1: Use mirroring
 - ◆ RAID Level 4, level 5: Parity based redundancy

Evaluation

- Capacity
 - ◆ N disks, B blocks per disk
 - ◆ $N*B$ blocks in total □ How much useful capacity is available to the clients of RAID?
- Reliability
 - ◆ How many disk faults can the RAID tolerate
- Performance
 - ◆ Read
 - ◆ write

RAID Level 0

- RAID Level 0 is the simplest form as **striping** blocks.
 - ◆ Spread the blocks across the disks in a round-robin fashion.

Disk 0	Disk 1	Disk 2	Disk 3
0	1	2	3
4	5	6	7
8	9	10	11
12	13	14	15

RAID-0: Simple Striping

RAID Level 0 (Cont.)

- Chunk size
 - ◆ Small chunk:
 - more intra-file parallelism
 - Larger positioning time: positioning time is the max positioning time of the disks
 - ◆ Large chunk:
 - Reduced intra-file parallelism
 - Smaller positioning time
- An example of RAID Label 0 with a bigger chunk size
 - ◆ Chunk size : 2 blocks (8KB)

Disk 0	Disk 1	Disk 2	Disk 3	
0	2	4	6	chunk size: 2 blocks
1	3	5	7	
5	10	12	14	
9	11	13	15	

Striping with a Bigger Chunk Size

RAID Level 0 Analysis

- ▣ Evaluate the capacity, reliability, performance of striping.
 - ◆ First way: single request latency
 - ▣ How much parallelism can exist during a single I/O operation.
 - ◆ Second way: steady-state throughput of the RAID:
 - ▣ Total bandwidth of many concurrent requests.

RAID Level 0 Analysis (Cont.)

Single Disk

- ♦ Average seek time: 7 ms
- ♦ Average rotational delay: 3 ms
- ♦ Transfer rate of disk: 50 MB/s

Single Disk Performance

- ♦ 10 Mbyte seq. IO, $S = \frac{\text{Amount of Data}}{\text{Time to access}} = \frac{10 \text{ MB}}{(7+3+200)=210 \text{ ms}} = 47.62 \text{ MB /s}$
- ♦ 10 Kbyte Random IO, $R = \frac{\text{Amount of Data}}{\text{Time to access}} = \frac{10 \text{ KB}}{(7+3+0.195)=10.195 \text{ ms}} = 0.981 \text{ MB /s}$

RAID 0

- ♦ Random write, random read = $N \cdot R$
- ♦ Sequential write, sequential read = $N \cdot S$

RAID Level 1

- RAID Level 1 is mirroring
 - ◆ Copy more than one of each block in the system.
 - ◆ Copy block places on a separate disk to tolerate the disk failures.

Disk 0	Disk 1	Disk 2	Disk 3
0	0	1	1
2	2	3	3
4	4	5	5
6	6	7	7

Simple RAID-1: Mirroring

RAID level 1

- Capacity $N*B/2$
- Reliability
 - ◆ From one to upto $N/2$ depending upon the failure disk
- Performance
 - ◆ Sequential write: $N*S/2$
 - ◆ Sequential read: $N*S/2$
 - ◆ Random write: $N*R/2$
 - ◆ Random Read: $N*R$

RAID Level 4

- RAID Level 4 is to add redundancy to a disk array as **parity**.

* P: Parity

Disk 0	Disk 1	Disk 2	Disk 3	Disk 4
0	1	2	3	P0
4	5	6	7	P1
8	9	10	11	P2
12	13	14	15	P3

Simple RAID-4 with parity

Disk 0	Disk 1	Disk 2	Disk 3	Disk 4
0	0	1	1	xor(0,0,1,1)
0	1	0	0	Xor(0,1,0,0)

RAID Level 4 (Cont.)

- The simple RAID Level 4 optimization known as a **Full-stripe write**.
 - ◆ Calculate the new value of P0 (Parity 0)
 - ◆ Write all of the blocks to the five disks above in parallel
 - ◆ Full-stripe writes are the most efficient way

Disk 0	Disk 1	Disk 2	Disk 3	Disk 4
0	1	2	3	P0
4	5	6	7	P1
8	9	10	11	P2
12	13	14	15	P3

Full-stripe Writes In RAID-4

Anlaysis

- Capacity: $(N-1)*B$
- Sequential read: $(N-1)*S$
- Sequential write: $(N-1)*S$ for full stripe write

Disk 0	Disk 1	Disk 2	Disk 3	Disk 4
0	1	2	3	P0
4	5	6	7	P1
8	9	10	11	P2
12	13	14	15	P3

- Random read: $(N-1)*R$

Analysis

□ Random write:

- ◆ Additive Parity update: read all blocks, update the block, compute the new parity and write the updated block and the updated parity.
- ◆ Subtractive parity update: read the parity, write (new xor old) xor (old parity). (read on parity disk)
- ◆ For each write, the RAID perform **4 physical I/O** . (two read and writes)

Disk 0	Disk 1	Disk 2	Disk 3	Disk 4
0	1	2	3	P0
*4	5	6	7	+P1
8	9	10	11	P2
12	*13	14	15	+P3

Random write performance: $(R/2)$ MB/sec

Small write problem happens

RAID Level 5

- RAID Level 5 is solution of small write problem.
 - ◆ small write problem cause parity-disk bottleneck of RAID Level 4.
 - ◆ works almost identically to RAID-4, except that it rotates the parity blocks across drives.
- RAID Level 5's Each stripe is now rotated across the disks.

Disk 0	Disk 1	Disk 2	Disk 3	Disk 4
0	1	2	3	P0
5	6	7	P1	4
10	11	P2	8	9
15	P3	12	13	14
P4	16	17	18	19

RAID-5 with Rotated Parity

Analysis

- Capacity: $(N-1)*B$
- Reliability: 1
- Performance
 - ◆ Sequential read, sequential write: $(N-1)S$
 - ◆ Random read: $N*R$
 - ◆ Random write: single write can cause 4 IO's (two read, two write), All N disks can work in parallel: $(N*R)/4$

Summary

	RAID-0	RAID-1	RAID-4	RAID-5
Capacity	N	N/2	N-1	N-1
Reliability	0	1 (for sure) N/2 (if lucky)	1	1
Throughput				
Sequential Read	NS	(N/2)S	(N-1)S	(N-1)S
Sequential Write	NS	(N/2) S	(N-1)S	(N-1)S
Random Read	NR	NR	(N-1)R	NR
Random Write	NR	(N/2)R	R/2	(N/4)R
Latency				
Read	D	D	D	D
Write	D	D	2D	2D